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Escola Superior de Tecnologia

Cycle in **Degree** Course
2nd Semester of 3rd Year
Course Unit: **Cloud Computing**
Lecturer: Osvaldo Santos

Practical work on Azure Serverless Computing

Objectives of this sheet

Initial contact with the Azure Function App service

Bibliography and useful links

Azure Functions documentation

<https://learn.microsoft.com/en-us/azure/azure-functions/functions-overview>

How to define CRON time expressions <https://crontab.guru/>



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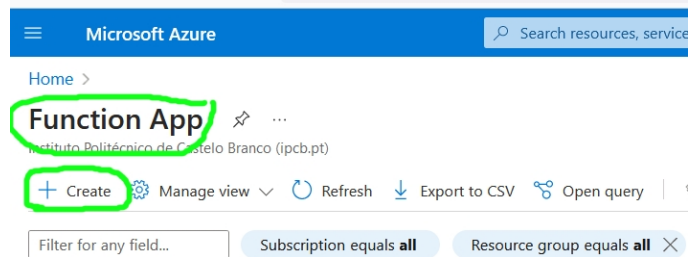
Practical work on Azure Serverless Computing

In this job you will use your ipcbcampus student account to access Azure

At the end of the job, you must delete all Azure resources so that you can preserve your balance

1) Create Function App

Before creating functions, you'll need to create the basis for running them, in Azure this is called **Function App**.



a) Create a new Function App, with the following characteristics:

i) Basics

- Resource Group: class
- Function App Name: "**your_name**"**random number**", e.g. vitor783457
- Publish: Code
- Runtime Stack: Node.js , Version 18 LTS
- Region: UK South
- Operating System: Linux
- Plan: Consumption

ii) Storage

- (New) accept the suggested name

iii) Networking

- Enable public access: On

iv) Monitoring

- Don't change anything

v) Deployment

- Don't change anything

vi) Tags



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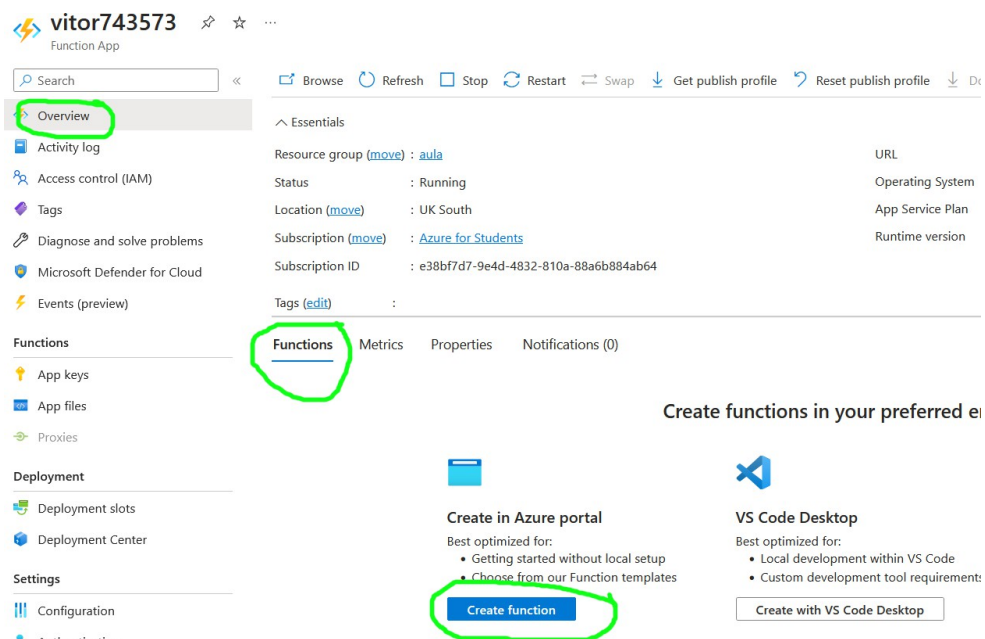
- Don't change anything

b) Create the Function App and wait for it to be created.

2) Run code periodically

Imagine you want to run some kind of task periodically (check the status of a database, check how many users are logged on, take a measurement, etc). One of the ways to do this without using a lot of computing resources is through Azure Functions, by creating a function **triggered by a timer**.

a) Inside the Function App you just created, create a new function.



b) Create a function with the name **TimerTrigger1**, to be edited in the portal, and to be called every 30 seconds. The definition of the time period uses the CRON syntax (see link on the first page).



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Create function



Select development environment

Instructions will vary based on your development environment. [Learn more](#)

Development environ...

Develop in portal

Select a template

Use a template to create a function. Triggers describe the type of events that invoke your functions. [Learn more](#)

Filter

Template	Description
HTTP trigger	A function that will be run whenever it receives an HTTP request, responding based on data in the body or query string
Timer trigger	A function that will be run on a specified schedule
Azure Queue Storage trigger	A function that will be run whenever a message is added to a specified Azure Storage queue
Azure Service Bus Queue trigger	A function that will be run whenever a message is added to a specified Service Bus queue
Azure Service Bus Topic trigger	A function that will be run whenever a message is added to the specified Service Bus topic
Azure Blob Storage trigger	A function that will be run whenever a blob is added to a specified container
Azure Event Hub trigger	A function that will be run whenever an event hub receives a new event

Template details

We need more information to create the Timer trigger function. [Learn more](#)

New Function*

TimerTrigger1

Schedule* ⓘ

*/*30 * * * * *

c) In the function code, you can edit several files (see figure below). What is the role of each of these 3 files?

TimerTrigger1 | Code + Test

Function

Search

Save Discard Refresh Test/Run Upload

Overview

Developer

Code + Test

Integration

Monitor

Function Keys

my-func67 \ TimerTrigger1 \

index.js

function.json

index.js

readme.md

1

Adding third party dependencies in the Azure portal is currently not supported



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- d) In the main code of the function (index.js), write the following Javascript code:

```
module.exports = async function (context, myTimer) { var
  timeStamp = new Date().toISOString();

  if (myTimer.isPastDue) {
    context.log(`JavaScript seems to be running slowly!`);
  }
  context.log(`Our function was called at this point in time: ${timeStamp}`);
};
```

Important note: sometimes, when copy-pasting PDF files, some errors occur with spaces, quotation marks and plicas. In Javascript, strings must be wrapped in backticks: ``` Check your code carefully and correct any errors.

- e) In the "Monitor" section, "Logs" tab, check that the logs that the function is writing every 30 seconds appear. You should see something like this:

Invocations Logs

Log Level ☐ Stop ☐ Copy ☐ Clear ☐ Open in Live Metrics ☐ Leave Feedback

```
Connected!
2024-03-14T14:53:30Z [Information] Executing 'Functions.TimerTrigger1' (Reason='Timer fired at 2024-03-14T14:53:30.0014872+00:00', Id=9747c45f-8778-4c07-966f-c5070e83f2c6)
2024-03-14T14:53:30Z [Verbose] Sending invocation id: '9747c45f-8778-4c07-966f-c5070e83f2c6
2024-03-14T14:53:30Z [Verbose] Posting invocation id:9747c45f-8778-4c07-966f-c5070e83f2c6 on workerId:2f9f20c3-69a4-43b2-8539-590840e9b6b2
2024-03-14T14:53:30Z [Information] Our function was called at this point in time: 2024-03-14T14:53:30.035Z
2024-03-14T14:53:30Z [Information] Executed 'Functions.TimerTrigger1' (Succeeded, Id=9747c45f-8778-4c07-966f-c5070e83f2c6, Duration=50ms)
2024-03-14T14:54:00Z [Information] Executing 'Functions.TimerTrigger1' (Reason='Timer fired at 2024-03-14T14:53:59.9999241+00:00', Id=b9d0330c-45c3-4d6b-af58-ec3403054079)
2024-03-14T14:54:00Z [Verbose] Sending invocation id: 'b9d0330c-45c3-4d6b-af58-ec3403054079
2024-03-14T14:54:00Z [Verbose] Posting invocation id:b9d0330c-45c3-4d6b-af58-ec3403054079 on workerId:2f9f20c3-69a4-43b2-8539-590840e9b6b2
2024-03-14T14:54:00Z [Information] Our function was called at this point in time: 2024-03-14T14:54:00.008Z
2024-03-14T14:54:00Z [Information] Executed 'Functions.TimerTrigger1' (Succeeded, Id=b9d0330c-45c3-4d6b-af58-ec3403054079, Duration=15ms)
2024-03-14T14:54:30Z [Information] Executing 'Functions.TimerTrigger1' (Reason='Timer fired at 2024-03-14T14:54:30.0006667+00:00', Id=60cf56dc-cd3b-4049-a091-a2522f9fd384)
2024-03-14T14:54:30Z [Verbose] Sending invocation id: '60cf56dc-cd3b-4049-a091-a2522f9fd384
2024-03-14T14:54:30Z [Verbose] Posting invocation id:60cf56dc-cd3b-4049-a091-a2522f9fd384 on workerId:2f9f20c3-69a4-43b2-8539-590840e9b6b2
2024-03-14T14:54:30Z [Information] Our function was called at this point in time: 2024-03-14T14:54:30.010Z
2024-03-14T14:54:30Z [Information] Executed 'Functions.TimerTrigger1' (Succeeded, Id=60cf56dc-cd3b-4049-a091-a2522f9fd384, Duration=14ms)
2024-03-14T14:55:00Z [Information] Executing 'Functions.TimerTrigger1' (Reason='Timer fired at 2024-03-14T14:55:00.0016243+00:00', Id=20777c95-039f-4464-b9ad-cf502fc85e95)
```



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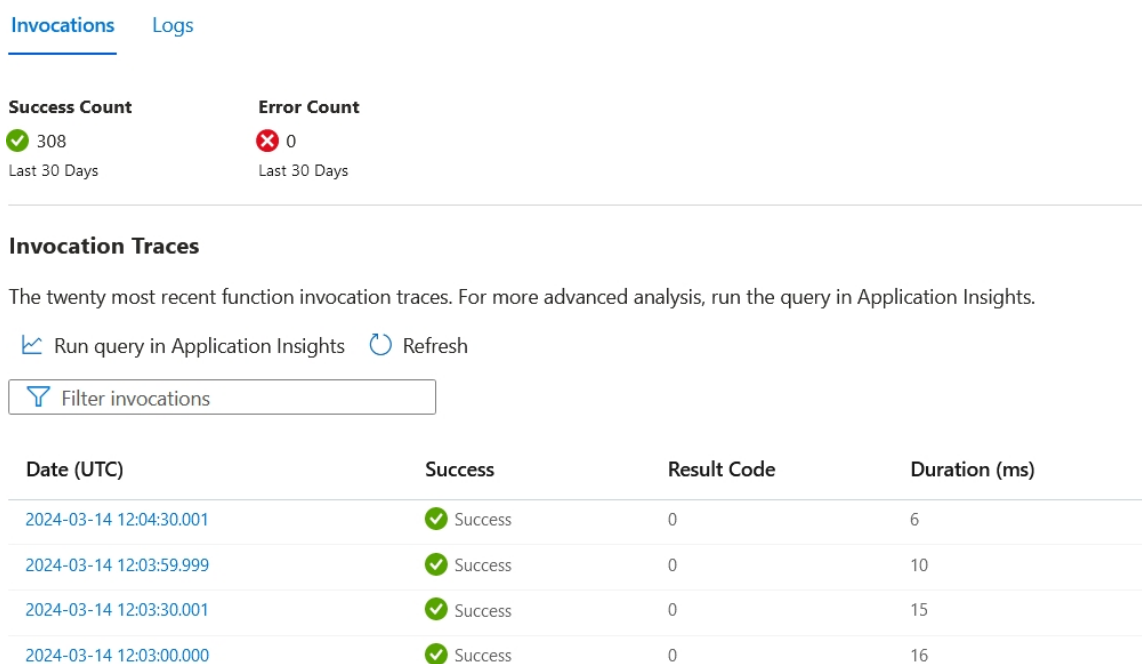


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- f) In the "Monitor" section, "Invocations" tab, check that the list of times the Function has been invoked. You should observe something like this:



- g) Is the function invoked every 30 seconds, as programmed?
- h) Check the average execution time of the function (how many ms of CPU time were used). If it runs every 30 seconds for a month, what will the total accumulated execution time be? How much will this function cost at the end of the month?

3) Creating a triggered function via HTTP

Imagine that you want to run some kind of task through a function. This task should be executed when, from the Internet, some other system invokes a certain URL via HTTP. Let's study how to implement something like this.

- a) Within the Function App, create a new function, with the name HTTPTrigger1, to be edited in the portal, and with authorization at the function level. What does function-level authorization mean?



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Development environ...

 Develop in portal

Select a template

Use a template to create a function. Triggers describe the type of events that invoke your functions. [Learn more](#)

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Azure Blob Storage trigger	A function that will be run whenever a blob is added to a specified container
Azure Event Hub trigger	A function that will be run whenever an event hub receives a new event

Template details

We need more information to create the HTTP trigger function. [Learn more](#)

New Function *

HttpTrigger1

Authorization level * ⓘ

Function

Practical work on Azure Serverless Computing

b) In the function code, write the following Javascript code:

```
module.exports = async function (context, req) {  
    context.log('The function has been called !');  
  
    const body = req.body;  
    if (body)  
        var bodyContent=JSON.stringify(body);  
  
    const responseMessage = body  
        ? `Data received: ${bodyContent}`  
        : `Please send data in the HTTP body`;  
  
    context.log(responseMessage);  
  
    context.res = {  
        // status: 200, /* Defaults to 200 */ body:  
        responseMessage  
    };  
}
```

Important note: sometimes, when copy-pasting PDF files, some errors occur with spaces, quotation marks and plicas. In Javascript, strings must be wrapped in backticks: ``` Check your code carefully and correct any errors.

c) Test the function using the "Test/Run" option.

d) Now get the URL from the "Get function URL" function (Function Key option) and save it in a text file. You should get something like this:

```
https://vitor98234.azurewebsites.net/api/HttpTrigger1?code=LH8ShkB0QssSUKm-  
Zl-WNoF1UNgbzdTdNmydVPYLZYGbAzFu6VM_HA==
```

e) What will that code parameter be used for?

f) Try opening the URL you obtained in a browser. What happens? Check in the Function Monitor that the function invocation has been logged.

g) Now try invoking the function from Postman, sending some JSON data.



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The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** `https://vitor98234.azurewebsites.net/api/HttpTrigger1?code=LH8ShkB0QssSUKm-ZI-WNoF1UNgbzdTdNmydVPYLZY...`
- Body (JSON):**

```
{  "id": "3984573",  "local": "Aveiro",  "Temperatura": 25}
```
- Status:** 200 OK
- Time:** 68 ms
- Size:** 289 B
- Response Body (Text):**

```
1 Foram recebidos estes dados: {"id":"3984573","local":"Aveiro","Temperatura":25}
```

- h) Was the function invoked correctly? Do the logs show the data that was sent?
- i) Could this type of function invoked remotely via HTTP be useful for IoT? Could they be useful for connecting Azure applications to other remote applications?
- j) Now try sending data to the function using the curl utility.

Tip:

```
curl -d '{"temperature': '26.8','humidity':'56%'}' "function URL"
```



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4) Use GitHub as a code repository for Azure functions

- a) In your GitHub account, create a repository with the source code of a timer-triggered function (every 30 seconds), called **timer7**, so that it only creates a record in the log whenever it is called.

Tip: if you can't do this part, you can use the code provided by the teacher in moodle

- b) Create a new Function App, linked to your GitHub account, and associate it with the GitHub repository you want to use.
just created, with "continuous deployment" activated.
- c) Check that the role has been created in Azure and that it works as expected.
- d) Now, on GitHub, change the periodicity of the function's execution to 20 seconds (edit the function.json file).
- e) Wait a few minutes for the changes to be implemented. The function now runs every 20 seconds ?

5) Use Visual Studio Code as a code editor

- a) Try creating a new time-triggered function, but now by creating the code through Visual Studio Code, and directly exporting the code to Azure from Visual Studio Code.

Create function

Select development environment

Instructions will vary based on your development environment. [Learn more](#)

Development environ...

 VS Code

6) Other types of Azure functions

- a) Read the Azure Functions documentation and see what other types of triggers (apart from timers and HTTP) can be used to trigger functions.



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You must now delete all Azure resources so that you can preserve your balance